

WeldCheck

Guided SMAW practice: set up, weld, review and improve

1 Choose exercise	2 Check setup	3 Weld + photo	4 Review + improve
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What WeldCheck is for

WeldCheck joins the approved exercise, actual setup, visible photograph result, correction plan and supervisor feedback in one practice record. It is a learning and record-keeping aid.

It does not certify a weld or determine internal condition, penetration, mechanical strength, safety or welding-code compliance.

1. Choose an approved exercise

- Select the exercise supplied by your workshop.
- Give the sample a number and optionally record the student, date and a short note.
- The exercise requirements come from a supervisor-approved template. If no exercise is shown, ask the administrator to add one. WeldCheck does not invent welding values.

2. Record and check the actual setup

- Record the plate thickness, electrode classification and size, welding position, and machine current actually being used.
- Complete each preparation check supplied with the exercise.
- WeldCheck shows each mismatch and the approved requirement. Correct or confirm every item before continuing.
- Ready means the recorded setup matches the exercise rules. It does not predict the finished weld quality.

3. Weld safely and add a clear photograph

- Follow workshop safety rules and supervisor instructions.
- After welding, let the sample cool and clean loose slag from the visible surface.
- Use even light. Keep the bead and both weld edges in view. Hold the camera steady.
- Brightness, contrast, sharpness and image size are checked locally. A poor photograph is stopped before AI review.

4. Read the visible result

- Acceptable appearance means none of WeldCheck's supported visible issue conditions was evident in the photograph.
- Visible issues may include visible porosity, undercut, excessive spatter or an irregular bead. More than one may be recorded.
- Retake photo means the image is not clear enough to assess.
- The AI describes visible evidence only. It does not use the recorded setup to guess a cause.

Review, improve and keep the record

Confirm what you can see

Choose Matches, Correct conditions, or Not sure. A correction is saved beside the original AI result and never overwrites it.

Plan the next attempt

Select what you will check or practise, add a short note, then start the next attempt. Setup details are copied for convenience, but the preparation checks must be completed again.

Compare attempts

After the next photograph, WeldCheck shows setup changes and which visible conditions were resolved, remain, or are new. This is a record of visible evidence, not proof of cause.

Supervisor review

A supervisor may record Accepted for this exercise, Another attempt needed, or Review incomplete, plus an optional name and comment. This field is a project record and is not an identity-verified digital signature.

Reports and downloads

Use Print or save PDF on an attempt for a readable report. History can download one CSV row per attempt and a full JSON backup. Photograph files are not included in exports.

Account and privacy

- Continue as a private guest, or connect Google to keep the same records across devices.
- Cloud photographs are stored privately under the account. Short-lived secure links display them.
- Keep CSV, JSON and printed reports private when they contain names or workshop records.

Important scope

Use qualified review and suitable tests for safety-critical decisions. WeldCheck covers the visible surface shown in one photograph only.